THE NOVA RESEARCH CURRICULUM

General Science and Social Studies

Grade 5 · Term 3

Student edition

General Science: SC-5.7 electricity and magnetism, SC-5.8 structure of the Earth and soil, SC-5.9 satellites, SC-5.10 independent fair-test report. Social Studies: SS-5.5 Greek, Roman and Gandhara civilisations; SS-5.6 economics and the State Bank of Pakistan.

General Science · lesson notes

- **SC-5.7 (Weeks 1–4):** static electricity (rub a balloon on hair and pick up paper); a simple circuit (cell, wires, bulb, switch); conductors and insulators; magnets have north and south poles, like poles repel and unlike poles attract. An electromagnet only with an adult. Safety: never touch mains sockets or wires.
- **SC-5.8 (Weeks 5–8):** the Earth's crust, mantle and core (an egg or peach model); soil types: sandy, clay, loam; which holds most water (fair test).
- **SC-5.9 (Weeks 9–10):** natural satellites (the Moon) and artificial satellites for weather, communication, navigation and mapping; Pakistan's PakSat-1R (communication) and PRSS-1 (remote sensing).
- **SC-5.10 (all term):** each child plans, carries out and reports one fair test alone, with a table, a bar graph and "I found that … / Next time I would …".

Social Studies · lesson notes

- **SS-5.5 (Weeks 1–4):** Greeks (city-states, early democracy in Athens, the Olympic Games), Romans (roads, laws, aqueducts), Gandhara (in today's northern Pakistan, around Taxila and the Peshawar valley; famous for Buddhist stone sculpture and learning). Visit Taxila Museum virtually.
- **SS-5.6 (Weeks 5–8):** public goods (roads, parks) vs private goods (a bicycle); inflation; imports and exports; entrepreneurship; the State Bank of Pakistan issues currency notes, oversees banks and manages the money supply.

▶ **Try it on NovaSchool** (open **Simulations** on your dashboard):

- **Conductors and insulators** — Put each object into the gap in the circuit. Does the bulb light?

Workbook pages

Page	Outcome	Activity
1	SC-5.7	Draw a circuit with a switch; conductor or insulator sort.
2	SC-5.8	Label the Earth's layers; soil-water fair test.
3	SC-5.9	Uses of satellites poster.
4	SC-5.10	My independent fair-test report.

Page	Outcome	Activity
5	SS-5.5	Compare three civilisations in a table.
6	SS-5.6	Public or private goods sort; what the State Bank does.

Question bank

#	Outcome	Question	Your answer
1	SC-5.7	Which is a good conductor of electricity?	plastic / rubber / copper / wood
2	SC-5.7	Like poles of magnets …	attract / repel / stick together / melt
3	SC-5.7	A switch opens and closes a circuit. True or false.	True / False
4	SC-5.8	The outer layer of the Earth is the …	core / mantle / crust / atmosphere
5	SC-5.8	Which soil is best for most crops?	sandy / clay / loam / rocky
6	SC-5.9	The Moon is a natural satellite. True or false.	True / False
7	SC-5.9	Name one use of artificial satellites.	_____
8	SC-5.10	A fair test changes only … thing at a time.	one / two / every / no
9	SS-5.5	The Olympic Games began in ancient …	Rome / Greece / Egypt / Gandhara
10	SS-5.5	Taxila was a centre of the Gandhara civilisation. True or false.	True / False
11	SS-5.5	The Romans were famous for building …	pyramids / roads and aqueducts / the Great Wall / Mohenjo-daro
12	SS-5.6	A public park is a … good.	private / public
13	SS-5.6	Which institution issues Pakistan's currency notes?	a commercial bank / the State Bank of Pakistan / a shop / the post office
14	SS-5.6	Goods sold to other countries are called …	imports / exports / taxes / loans